

Coded Beam Training

Beam 0 = 0°

1 = 5.6°

2

3

...

43 = 240° (UE)

...

64 = 360°

Paper: 10 Wide beams

1WB = 32 Narrow beams
power

$$2^n \text{ combinations} = 2^6 = 64$$

$$43 = 101011 \text{ (6 bits)}$$

$$\text{No of bits} = \log_2(64) = 6$$

000000(0), 000001(1), ..., 111111(63)

64 antennas → 64 beams (6 bits)
(narrow/wide)

Narrow beam (high SNR)

laser pointer
long distance

Slow + Fail

wrong (x)
angle



BS

2km

Beam #43



UE

noise
+
low power

Poor sig
power
(-15dB)

Wide beam (multi lobe)

wide area / more area
short distance

approx UE angle

(low SNR)

Hamming code → error correction
6 bits → 10 bits
+4 (parity bits)

Antennas = 64
Coverage = 360°
user angle = 240°
(43 beam)

Power

10 bits

0001011101
9 8 7 6 5 4 3 2 1 0 MSB ↑ error ↓
LSB 0.5% 50%

MSB → more power LSB → less power

64 Beams (codebook)

BS sends 10 widebeams to UE, UE gives 10 bit feedback (43 is best)

Decoder → (removes noise) from received bits

Hamming decoder

Gains

$$G_{\text{narrow}} = 10 \log_{10}(64) = 18 \text{ dB}$$

$$G_{\text{wide}} = G_{\text{narrow}} - 10 \log_{10}(2) = 15.05 \text{ dB (32 slice)}$$

↓

32 narrow
slices

$$18.06 - 3.01 = 15.05 \text{ dB}$$

($\because$ Beamwidth
doubles) ↓

Power density
halves

↓

So $10 \log_{10}(2)$

= 3 dB (loss)

Total $\rightarrow$ 64 narrow beams (360° circle)

D1 $\rightarrow$ 0-31 (32 beams) $\rightarrow 180^\circ$ Half circle

D2 $\rightarrow$ 32-63 (32 beams) Half circle

D3 $\rightarrow$

⋮

D10 $\rightarrow$

$\frac{64}{4} = 16$ beams per 90° quadrant
↓
4 quadrants

Power Allocation

MSB is given more power $\rightarrow$ to reduce errors

LSB less power

We cannot give more power to all bits because power is limited resource.